Sports Med
DOI 10.1007/s40279-017-0783-2

REVIEW ARTICLE

A Narrative Review of the Physical Demands and Injury Incidence in American Football: Application of Current Knowledge and Practices in Workload Management

Toby Edwards[1] · Tania Spiteri[1] · Benjamin Piggott[1] · G. Gregory Haff[2] · Christopher Joyce[1]

© Springer International Publishing AG 2017

Abstract The sport of American football (AmF) exposes athletes to high-velocity movements and frequent collisions during competition and training, placing them at risk of contact and non-contact injury. Due to the combative nature of the game, the majority of injuries are caused by player contact; however, a significant number are also non-contact soft-tissue injuries. The literature suggests that this mechanism of injury can be prevented through workload monitoring and management. The recent introduction of microtechnology into AmF allows practitioners and coaches to quantify the external workload of training and competition to further understand the demands of the sport. Significant workload differences exist between positions during training and competition; coupling this with large differences in anthropometric and physical characteristics between and within positions suggests that the training response and physiological adaptations will be highly individual. Effective athlete monitoring and management allows practitioners and coaches to identify how athletes are coping with the prescribed training load and, subsequently, if they are prepared for competition. Several evidence-based principles exist that can be adapted and applied to AmF and could decrease the risk of injury and optimise athletic performance.

Key Points

In American football (AmF), non-lineman, particularly the wide receiver, defensive back and linebacker, position groups perform at higher movement velocities, whilst running backs endure the greatest number of severe collisions. Microtechnology allows practitioners to build the athletes' ability to tolerate collisions and high movement velocities in competition.

Soft-tissue injuries are the second highest category of injury in AmF, with the major proportion occurring after breaks in training such as pre-season training camp. Practitioners should acknowledge the training history and previous workload of athletes when prescribing training during these periods.

High average inertial load values generated by performing explosive acceleration, deceleration and change of direction decrease the risk of injury in AmF athletes. Establishing and maintaining an appropriate level of neuromuscular control and responsiveness should be a focus for practitioners.

✉ Toby Edwards
toby.edwards1@my.nd.edu.au

1 School of Health Sciences, University of Notre Dame Australia, 33 Phillimore Street, Fremantle, WA 6959, Australia

2 Centre for Exercise and Sport Science Research, Edith Cowan University, Perth, WA, Australia

1 Introduction

The modern-day game of American football (AmF) is primarily comprised of repeated high-intensity bouts involving change of direction, acceleration, deceleration and sprinting, followed by longer periods of low-intensity

Published online: 25 September 2017

rest, recovery and tactical play [1–3]. Athletes also experience high contact loads through tackling and blocking [2], with the resultant collision loads greatly differing depending on positional requirements [2, 3]. For ease of clarification, collisions are defined as direct contact with another player in the instance of tackling or blocking. Impacts are defined as ground-reaction forces generated by activities such as running, jumping and change of direction. Both collisions and impacts result in inertial load sustained by the body [4–9]. The high-intensity nature of the game combined with the frequent collisions places AmF athletes at a considerable risk of contact and non-contact injury [10]. One method to address the injury concerns is through appropriate workload management. This area of research has received increased interest from practitioners and researchers of the elite team sport population [11–22]; however, there is a lack of literature investigating this issue in AmF. This is primarily due to the relatively recent introduction of player tracking technology to quantify external demands of training and competition and the feasibility of acquiring enough units to monitor athletes. Few studies have reported the physical demands of training [23] and competition [2, 3], with only one published study investigating the link between workload and injury risk in AmF [22]. This study reported that high inertial loads decreased the risk of either contact or non-contact musculoskeletal strains and sprains. Athletes who have superior neuromuscular control and responsiveness have the ability to produce extremely rapid changes of direction, acceleration and deceleration, and therefore generate high inertial load values, which may offer a protective effect [22]. Current research suggests that this is one of several factors that need to be considered when monitoring athletes [14, 24, 25]. Further understanding of how workload relates to injury risk in AmF through either positive (increased physical capacity) or negative (injury, illness, overtraining) training adaptations is vital in order to maximise athletic performance and player availability.

In order to bridge the gap in research and investigate the link between workload and injury, several steps need to be taken. These include an increased investment from the athletic organisation into sports performance, a change in paradigm that accepts and embraces modern sport science initiatives, and greater knowledge and understanding of athlete monitoring and recovery protocols. Collectively, this requires a multidisciplinary approach between coaches, athletes, strength and conditioning staff, sport scientists, athletic trainers and medical staff as each role is somewhat involved in the training process [26]. The objective of this review is to discuss the current literature surrounding the physical demands and injury incidence of AmF and provide a discussion on how current knowledge and practices in training load management can be applied to AmF to reduce the risk of injury and optimise performance.

2 American Football (AmF) Positions

A unique aspect of AmF is that each position on the field has specific responsibilities and vastly different anthropometric and physical characteristics [1, 27, 28]. A basic graphical depiction of each position on the field is provided in Fig. 1. Whilst there are some inconsistencies with positional clarification, similar positions are often classified together based on similar attributes or technical skill requirements. For consistency between the articles reviewed, positions were defined and grouped as the offensive tackle (OT), offensive guard (OG), tight end (TE) and centre (C) positions, commonly known as the offensive line (OL); and the defensive end (DE) and defensive tackle (DT) positions are referred to as the defensive line (DL). The positions that make up the offensive and DLs can also be grouped together and called 'lineman', whilst the wide receiver (WR), quarterback (QB), running back (RB), defensive back (DB) and linebackers (LB) are often represented as 'non-lineman'. Table 1 summarises the anthropometric and strength characteristics of each position to highlight the distinct differences between individual position groups and between the lineman and non-lineman position groups in National Collegiate Athletic Association (NCAA) Division I AmF [28]. When comparing these attributes across teams it is important to consider the style of play the head coach adopts as this may greatly influence both the physical and anthropometric characteristics of positions. The different position-specific characteristics suggest that the training prescription should be individualised to meet the needs of each position as the balance between external physical demands and internal responses varies according to a multitude of factors including, but not limited to, anaerobic and aerobic fitness, neuromuscular control, age and the training stimulus [25, 29].

3 Physical Demands of AmF

3.1 Microtechnology

Throughout the last decade microtechnology has become commonplace in sport and sport research for monitoring the physical demands of athletes during training and competition in soccer, Australian rules football (ARF), rugby league and rugby union [4, 7]. The integration of micro-inertial sensors such as global positioning systems (GPS), triaxial accelerometers, gyroscopes and magnetometers into a single microelectromechanical system

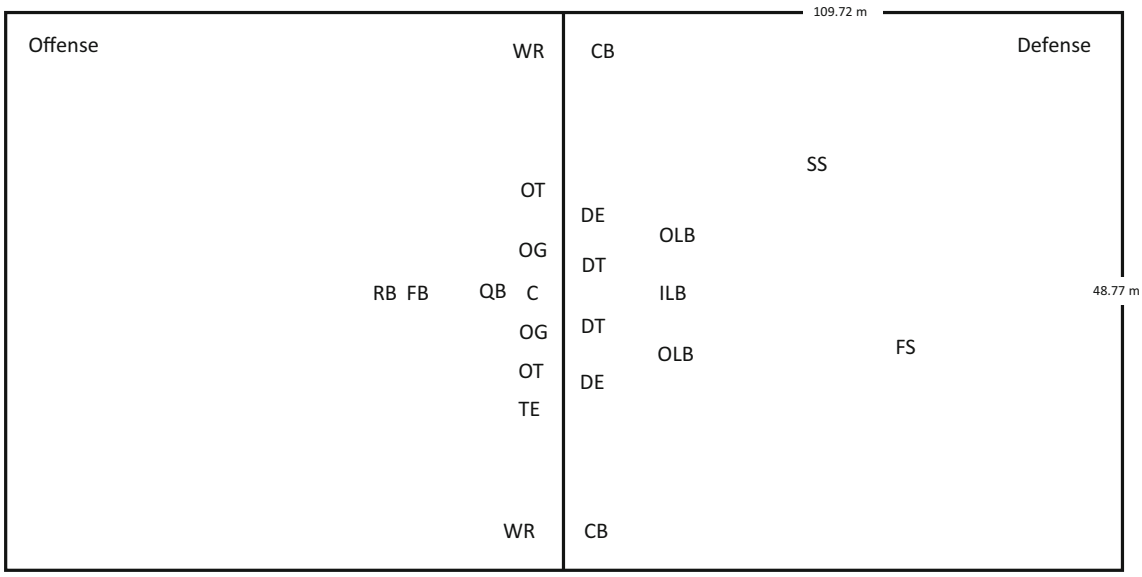

Fig. 1 Common offense and defense formation (not to scale). Offense: *C* centre, *FB* full back, *OG* offensive guard, *OT* offensive tackle *QB* quarterback, *RB* running back, *TE* tight end, *WR* wide receiver. Defense: *CB* corner back, *DE* defensive end, *FS* free safety, *DT* defensive tackle, *ILB* inside linebacker, *OLB* outside linebacker, *SS* strong safety

Table 1 Summary of position-specific anthropometric and physical performance qualities of Division I collegiate American football [9]

Position	Anthropometric tests			Performance tests			
	Height (cm)	Weight (kg)	Body fat %	Squat (kg)	Bench press (kg)	Vertical jump (cm)	40-yard sprint (s)
Non-lineman							
WR	183.0 ± 2.3	85.6 ± 7.0	6.6 ± 2.2	205.6 ± 40.4	151.2 ± 26.5	87.4 ± 7.0	4.48 ± 0.1
DB	180.1 ± 1.9	85.1 ± 5.8	6.9 ± 2.1	207.8 ± 39.6	142.4 ± 16.6	87.8 ± 7.8	4.52 ± 0.2
LB	185.1 ± 1.4	103.8 ± 5.2	9.5 ± 3.0	240.5 ± 36.5	159.5 ± 23.7	83.2 ± 7.8	4.64 ± 0.2
RB	181.3 ± 1.5	97.3 ± 8.9	8.8 ± 2.9	233.3 ± 32.6	174.9 ± 23.7	85.9 ± 7.7	4.53 ± 0.2
QB	185.8 ± 2.0	92.7 ± 6.3	9.2 ± 2.2	200.2 ± 45.0	162.9 ± 21.7	80.7 ± 6.4	4.7 ± 0.1
Lineman							
DL	188.0 ± 2.7	120.7 ± 8.8	14.1 ± 3.8	246.5 ± 34.8	180.1 ± 24.0	77.9 ± 8.2	4.85 ± 0.2
OL	191.3 ± 1.9	133.2 ± 8.1	19.2 ± 2.8	251.3 ± 33.8	174.0 ± 27.6	68.8 ± 6.2	5.12 ± 0.2
TE	190.3 ± 1.1	113.7 ± 6.4	12.6 ± 2.9	232.4 ± 37.3	172.4 ± 16.5	79.6 ± 7.2	4.78 ± 0.2

Data are given as mean ± standard deviation

DB defensive back, *DL* defensive line, *LB* linebacker, *OL* offensive line, *QB* quarterback, *RB* running back, *TE* tight end, *WR* wide receiver

(MEM) provides practitioners with a wide array of data to be collected and used to inform decisions on athlete training and performance [4, 7]. These metrics include distance, high-speed running, accelerations, decelerations and collisions. The accuracy of MEMs to measure movement patterns relies heavily on the sampling frequency of the sensors, with higher sampling rates proving to be more valid and reliable at detecting movement patterns at low and high speeds [4, 7]. Triaxial accelerometers, gyroscopes and magnetometers have a much higher sampling frequency (100 Hz) than GPS (5–20 Hz) and provide a valid and reliable method to detect collisions and not just movement patterns [15, 30, 31]. Research using microtechnology devices in sport have primarily focused on quantifying external load using total distance and high-speed running collected via GPS [7]. When primary interest relates to acceleration, deceleration and collisions, the devices are more readily known as inertial measurement units (IMUs). Research using IMUs to investigate impacts and collisions in rugby league, rugby union and AmF have used the impact classification guide seen in Table 2, which is based on manufacturer recommendations

Table 2 Impact classification zones

Zone	Gravitational force (G-force)	Collision classification
1	<5.0–6.0	Hard acceleration, deceleration or change of direction whilst running
2	6.1–6.5	Minor collision with opponent, contact with ground
3	6.6–7.0	Making a tackle or being tackled at a moderate velocity
4	7.1–8.0	High-intensity collision with opponent, making direct front-on tackle on opponent travelling at moderate velocity, being tackled by multiple opposition players when running at submaximal velocity
5	8.1–10.0	High-intensity collision with opponent, making direct front-on tackle on opponent travelling at high velocity, being tackled by multiple opposition players when running at near-maximal velocity
6	>10.0	High-intensity collision with opponent, making direct front-on tackle on opponent traveling at high velocity, being tackled by multiple opposition players when running at maximal velocity

(GPSports, Canberra, ACT, Australia) [2, 32–34]. The higher sampling frequency and high-speed frequent collision nature of AmF suggest that IMUs could be used to monitor external demands of AmF athletes through training and competition [2, 4, 7, 22].

3.2 Movement Profiles

To the authors' knowledge, only one study has objectively quantified the movement profiles of competition in AmF using GPS [3]. Table 3 displays a summary of the investigation and highlights the position-specific movement profiles of collegiate AmF players [3]. It was reported that the WR position covered significantly more moderate-intensity, high-intensity and sprinting distance than all other offensive position groups. The WR position also performed significantly more high-intensity movements such as sprint efforts and moderate-, high- and maximal-intensity accelerations and decelerations than the other offensive position group [3]. In regards to the defensive positions, the DB and LB position covered significantly more distance in all movement velocity zones than all other defensive position groups. The DB position also performed significantly more sprint efforts and moderate-, high- and maximal-intensity accelerations and decelerations than the DE and DT position groups [3]. Similar observations were reported earlier with non-lineman positions covering a higher percentage of total distance above 16.0 km/h in team drills, position drills and overall practice than lineman [23]. This suggests non-lineman, particularly the WR, DB and LB positions, perform at higher movement velocities than all the other position groups at training and in competition. The scarcity of comparable literature in AmF makes it difficult to yield further comparisons and therefore more research using GPS in AmF is needed in this area to achieve a greater consensus on positional movement

Table 3 Summary of position-specific movement profiles per game in Division I collegiate American football [5]

Position	Low-intensity distance (0–10 km/h) (m)	Moderate-intensity distance (10.1–16 km/h) (m)	High-intensity distance (16.1–23 km/h) (m)	Sprinting distance (>23 km/h) (m)	Total distance (m)	Average maximal speed (km/h)
Non-lineman						
WR	3546.2 ± 756.2	1530.9 ± 341.2	655.2 ± 196.3	315.8 ± 163.2	5530.6 ± 996.5	31.5 ± 2.2
DB	3448.7 ± 923.0	926.1 ± 247.4	513.8 ± 155.5	247.0 ± 113.1	4696.2 ± 1114.8	31.1 ± 1.9
LB	2989.1 ± 721.5	912.5 ± 271.4	435.0 ± 165.0	196.7 ± 104.7	4145.4 ± 980.3	29.6 ± 1.2
RB	2291.3 ± 482.0	738.4 ± 247.2	303.1 ± 118.7	101.2 ± 71.7	3140.6 ± 685.6	28.8 ± 2.5
QB	3661.5 ± 642.2	568.3 ± 147.8	138.1 ± 65.1	76.9 ± 46.0	3751.9 ± 801.9	29.4 ± 8.5
Lineman						
DE	2662.8 ± 652.5	665.2 ± 224.0	226.0 ± 96.1	29.2 ± 24.1	3276 ± 815.2	26.1 ± 2.6
TE	2579.2 ± 663.8	947.2 ± 155.5	336.5 ± 137.8	40.3 ± 47.4	3574.2 ± 603.0	25.3 ± 7.8
OL	2885.4 ± 663.8	913.2 ± 147.8	131.1 ± 65.7	9.3 ± 11.3	3652.4 ± 603.0	23.7 ± 2.8
DT	2499.5 ± 456.9	629.0 ± 249.0	158.6 ± 62.0	7.7 ± 10.9	3013.0 ± 650.9	23.5 ± 1.7

Data are given as mean ± standard deviation

DB defensive back, *DE* defensive end, *DT* defensive tackle, *LB* linebacker, *OL* offensive line, *QB* quarterback, *RB* running back, *TE* tight end, *WR* wide receiver

profiles, which could then benefit practitioners through appropriate training prescription.

3.3 Impact Profiles

In addition to the various movement demands, AmF athletes are also exposed to repeated high-contact collisions through blocking and tackling [2]. Position-specific impact profiles are summarised in Table 4, which highlight positional differences in the magnitude and frequency of impacts and collisions. Specifically, WRs are exposed to a greater magnitude of 'very light' (5.0–6.0 G-force [G]) and 'light to moderate' (6.1–6.5 G) impacts than other offensive position groups [2]. This is most likely a reflection of the movement profile of the WR position group as low-intensity impacts (2.0–6.0 G) can be attributed to walking and running, which leads to a large amount of 'very light' to 'moderate' impacts. The WR position group is also exposed to a large number of 'very heavy' (8.1–10.0 G) and 'severe' (>10.0 G) impacts, which most likely occur when being tackled at high velocity. However, RBs endure the greatest number of severe (>10 G) impacts during competition on the offensive team [2]. This may reflect the tactical demands of being the primary ball carrier on the offensive team leading to high-magnitude collisions with opposing players on the defensive team. In regards to the defensive positions, the DB, DE and LB position experience more 'very light' impacts than DTs; however, they are also exposed to a large number of 'severe' impacts. Meanwhile, DTs were involved in more 'moderate to heavy' (6.6–7.0 G), 'heavy' (7.1–8.0 G) and 'very heavy' (8.7–10.0 G) impacts than the other defensive positional group [2]. The quantification of collisions and impacts in competition may help coaches prescribe training to ensure athletes have the ability to tolerate physical collisions in competition [5]. Future work is required to differentiate impacts caused by movement such as jumping, landing and change of direction, and those caused by player collisions.

3.4 Work:Rest Ratios

A team's or coach's style of play can greatly influence the physical demands and speed of the game, which may also have a carry-over effect to the anthropometric and physical qualities of athletes in certain positions [35]. The average duration of all pass and run plays in collegiate AmF have been reported as 5.60 ± 1.71 and 4.86 ± 1.41 s, respectively [35]. The average duration of all plays determined by a team's style of play for run, pass and balanced offenses were described as 4.84 ± 1.37, 5.41 ± 1.56 and 5.44 ± 1.81 s, respectively [35]. The average work:rest ratio was 1:7; however, for tactical and technical purposes a team may run an up-tempo or slower-style offense to maximise the teams' strengths or take advantage of the opposition or current situation of the game. An up-tempo offense has a work:rest ratio of 1:3 compared with that of a slower-type offense of 1:10 [35]. The primary energy system utilised in most plays seems to be the adenosine triphosphate phosphocreatine system. However, it is important that the sporting coaches and practitioners communicate to ensure that athletes are prepared for the style of game the coach wants to adopt. Athletes need to be prepared adequately to perform in any situation.

The combination of information on movement demands, magnitude and frequency of collisions, and work:rest ratios in AmF may provide a more global assessment of the physical load (intensity and volume) that athletes are exposed to during competition. Future work is required to

Table 4 Summary of position-specific impact profiles per game in Division I collegiate American football [4]

Position	Very light (5.0–6.0 G)	Light to moderate (6.1–6.5 G)	Moderate to heavy (6.6–7.0 G)	Heavy (7.1–8.0 G)	Very heavy (8.1–10.0 G)	Severe (>10 G)
Non-lineman						
WR	4093 ± 791.6	1155.9 ± 401.7	172.7 ± 56.7	38.4 ± 14.7	11.1 ± 5.5	12.3 ± 5.0
DB	2938.9 ± 569.1	581.5 ± 186.6	100.9 ± 42.5	19.3 ± 9.5	7.4 ± 4.1	9.6 ± 4.9
LB	2638.9 ± 566.4	545.8 ± 287.3	100.1 ± 47.5	23.6 ± 13.3	9.3 ± 6.0	12.7 ± 7.4
RB	1929.9 ± 469.2	582.7 ± 184.8	78.4 ± 31.4	21.4 ± 10.4	9.5 ± 4.9	16.6 ± 7.9
QB	2060.7 ± 725.7	333.3 ± 109.9	44.3 ± 11.3	15.5 ± 5.3	9.3 ± 5.7	13.6 ± 5.9
Lineman						
DE	2319.1 ± 682.5	532.9 ± 202.2	105.9 ± 50.4	31.4 ± 17.2	11.7 ± 5.8	13.2 ± 6.9
TE	2615.3 ± 725.7	869.5 ± 255.6	175.2 ± 58.4	31.6 ± 14.5	9.1 ± 4.4	5.9 ± 2.3
OL	2372.8 ± 415.4	851.6 ± 222.9	162.1 ± 103.9	35.9 ± 18.7	12.9 ± 6.8	11.5 ± 5.9
DT	1847.4 ± 431.1	699.2 ± 215.6	198.4 ± 102.4	49.6 ± 20.9	18.0 ± 10.2	10.6 ± 4.6

Data are given as mean ± standard deviation

DB defensive back, *DE* defensive end, *DT* defensive tackle, *G* G-force, *LB* linebacker, *OL* offensive line, *QB* quarterback, *RB* running back, *TE* tight end, *WR* wide receiver

identify the extremes of competition demands (highest load scenarios) such as total high-speed running, maximum-intensity acceleration/deceleration and maximum speed in order for coaches and practitioners to facilitate training prescription to ensure athletes are adequately prepared for these situations. In addition, if the use of microtechnology continues to increase in AmF, more work needs to be conducted to identify impact classification and speed thresholds as the manufacturer recommendations may not be suitable for athlete monitoring purposes in AmF.

4 Injuries in AmF

4.1 Injury Rates

The high-intensity movement and collision nature of AmF places athletes at a susceptible risk of contact and non-contact injuries. Injury rates increase with the level of competition, as calculated per athletic exposure (AE) [36, 37]. An AE has been classified as one athlete participating in one practice or game, in an attempt to control for the varying number of training sessions and games that occur yearly [37]. Injury rates of 2.56/1000 AEs and 5.77/1000 AEs have been reported during high school and collegiate practices, respectively [37]. Reported injury rates during professional AmF training camps are 13.36/1000 AEs, more than double the injury rate in collegiate football [36]. At the collegiate level, injury rates are reportedly nine times higher during games than in-season practices (35.9 vs. 3.8/1000 AEs) [10]. Increased intensity of games can result in a greater number and magnitude of collisions, and subsequently a greater risk of injury than in practice sessions, making these reported data a logical finding. Seasonal changes in training focus during the collegiate year may vary the amount of external load athletes are exposed to. For example, during the spring season, practice often involves greater movement demands and an increased amount of contact work than fall (autumn) practice as the players do not need to prepare for weekly games [10]. The seasonal changes in training focus have been found to influence the player injury rate [10], with more than two times higher injury incidences in the spring than in fall practice (10 vs. 4/1000 AEs). During spring season practice, collegiate AmF coaches may influence injury rates as they can manipulate the prescription of the amount and intensity of player contact [10]. This is also an important time to build athletes' ability to tolerate physical collisions prior to the competitive season. As previously stated, microtechnology such as triaxial accelerometers in GPS units may help quantify the frequency and magnitude of contact during training sessions and competition. This could assist coaches in prescribing future training sessions in order to reduce injury rates and prepare players for competition.

4.2 Mechanism of Injury

When investigating the primary mechanism of injuries in collegiate AmF, a large percentage resulted from player contact [10]. Specifically, 78%, 69% and 57% of all injuries during games, spring practice and fall practice, respectively, were caused by player contact [10]. This is in support of more recent research reporting 72.1% of all game injuries were caused by direct player contact [38]. The most common injuries caused by player contact are soft-tissue knee injuries such as partial or complete tears of the menisci and/or cruciate ligament rupture (17.8%), ankle ligament sprains (15.6%) and concussions (6.8%) [10]. There is level 1 evidence that suggests alterations in the rules of contact sports can result in a significant decrease in the level of contact that can be achieved [39, 40]. Without rule changes it is practically impossible to decrease the number of player contacts in competition due to the nature of the game. The number of contacts could be monitored during practices and limited on days when athletes are fatigued, with the aim of reducing the risk and occurrence of injuries due to direct player contact.

Non-contact soft-tissue injuries such as musculotendon strains are the second highest mechanism of injury in collegiate AmF [10]. Specifically, 8.9%, 21.8% and 28.8% of all injuries during games, spring practice and fall practice are non-contact injuries, respectively [10]. The proportion of non-contact injuries is unclear, with recent research reporting that non-contact injuries represent 40% of all injuries at the collegiate level of competition [38]. Musculotendon strains have been considered as errors in training load prescription and are often viewed as preventable [13]. In the 5-week training camp period, muscle strains represent between 33% and 47% of all injuries while sprains represented between 33% and 44% [36]. Cumulatively, the training camp phase results in a large acute training stimulus aimed at developing athletes' physical capacities (strength, power, speed, agility), position-specific drills and restricted-contact scrimmages to improve technical ability [36]. It could be speculated that the high percentage of muscle strains during the training camp period could potentially be caused by the suspected acute training stimulus athletes are exposed to. In this instance, assessing prior training load history is important as the break between the end of season and preseason camp generally results in a low training base or chronic workload. A number of primary preventive measures could be employed by coaches, athletes and practitioners to decrease risk of injury, including completing more training prior to preseason camp, less training at the camp or a combination

of both to ensure training prescription is individualised [24]. This may create an environment in which training is more effective through prescribing sufficient intensity that allows favourable adaptations and decrease risk of injury and maladaptation [24].

5 The Workload–Injury Aetiology Model

Several multifactorial models of injury causation have attempted to identify the intrinsic (athlete-related) and extrinsic (activity-related) risk factors that can modify the possibility of an athlete attaining an injury [25, 41]. Within the updated workload–injury aetiology model, an athlete is predisposed to injury through modifiable risk factors such as aerobic capacity, strength and neuromuscular control as well as non-modifiable risk factors such as age, gender and anatomy [25]. This separates the risk factors into those that are susceptible to change through physiological adaptation and those that are constant. The inclusion of workloads within this model better explain the association between load and injury. The workload–injury model in Fig. 2 describes the application of workload as a 'vehicle' where athletes are exposed to external risk factors such as equipment and the environment, and potential inciting events such as cumulative tissue overload, collisions, falls and non-contact acute events [25]. This understanding of workload explains that it is not the direct cause of injury; however, it does contribute to injury risk through positive (increased aerobic capacity, strength) and negative (decreased neuromuscular control) training effects on modifiable internal risk factors, and also exposing the athlete to injurious situations [25].

According to the workload–injury aetiology model, the relationship between load and injury exists under three main conditions. The first is that excessive workloads increases athletes' exposure to external risk factors and potential inciting events, therefore increasing injury risk [25]. The second condition surrounds the rationale that negative changes to modifiable internal risk factors induced by workloads increase injury risk [25]. Finally, the third condition of the workload–injury aetiology model is that maximising positive changes to modifiable internal risk factors whilst minimising negative fatigue effects through exposing athletes to appropriate workloads will make athletes more resistant to injury [25]. The physical and

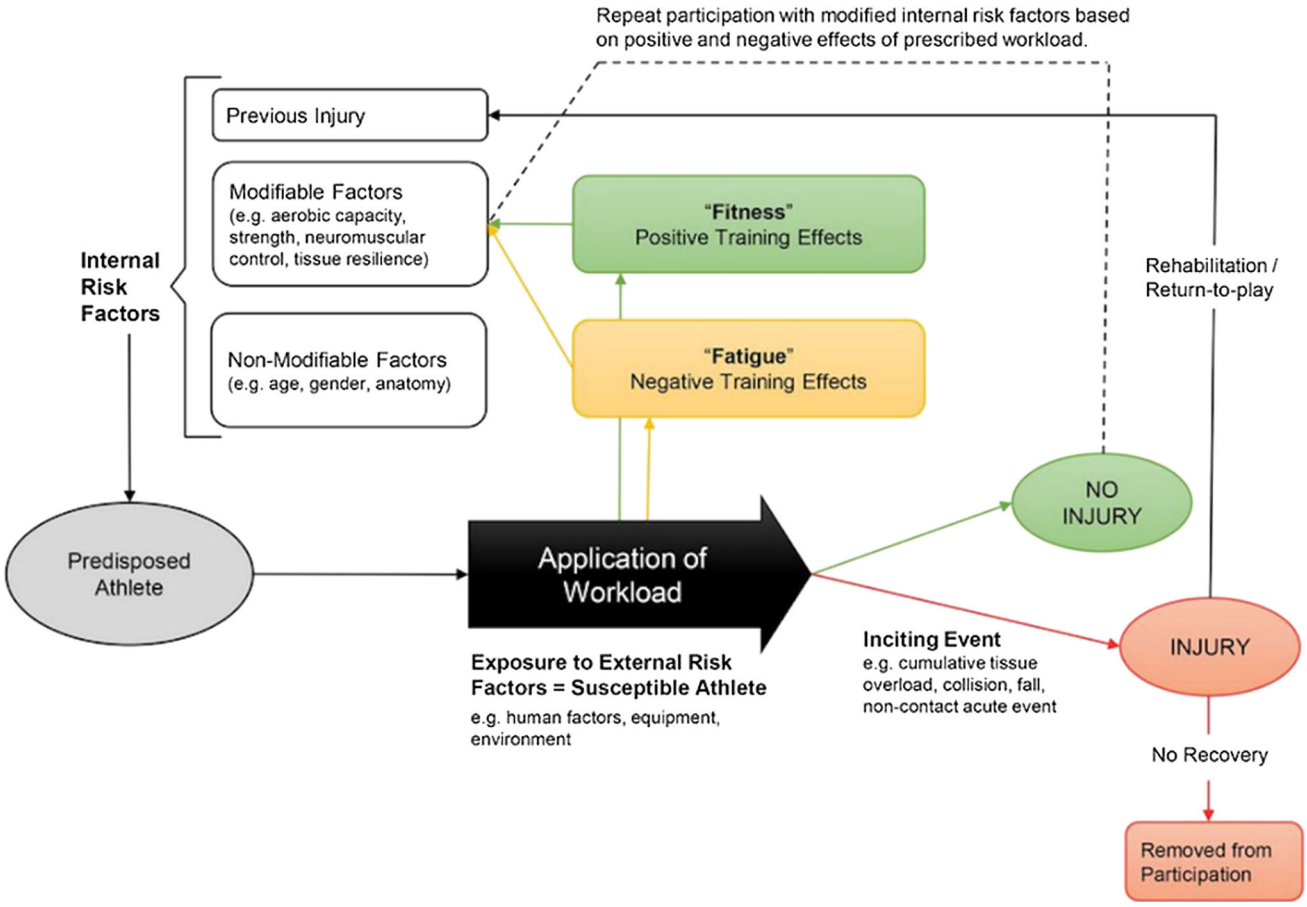

Fig. 2 The workload–injury aetiology model. Reproduced from Windt and Gabbett [25], with permission from BMJ Publishing Group Ltd

anthropometric qualities of AmF athletes are either modifiable or non-modifiable internal risk factors in the workload–aetiology model that impact the internal load placed on the body, thus predisposing the athlete to injury [25, 29, 42, 43]. As previously discussed, these qualities vary significantly between position groups [27, 28], suggesting that in order to maximise positive adaptations and reduce fatigue effects the training prescription must be managed accordingly as the response to workload is highly individual [29, 43]. However, there is a dearth of research investigating the relationship between individual internal risk factors on injury and workload in AmF compared with other collision sports. In ARF it was reported that players with superior aerobic capacity measured via a 2-km time trial rated training sessions of the same distance easier than players with an inferior aerobic capacity [43]. In rugby league players, decreases in aerobic and muscular power as well as an increase in skinfold measurements towards the end of a season resulted in increased injury rates [42]. Collectively, these findings in addition to previous knowledge on the highly position-specific anthropometric and physical characteristics of AmF suggest that longitudinal management and monitoring of individual characteristics, training load and training response need to be implemented.

6 Monitoring Workload in AmF

The high-intensity and high-impact nature of AmF make it difficult to prevent all injuries; however, a substantial proportion of injuries are non-contact soft-tissue injuries [10, 36–38]. This type of injury could be attributed to training load errors, which can be largely preventable through monitoring and managing individual athlete workloads [12, 14, 16–21, 24, 44–46]. Practitioners working with AmF need to expose athletes to a high number of collisions in training to better prepare players for the demands of competition [2, 6]. This is also true of positions who are frequently exposed to maximum-velocity accelerations, deceleration and running. However, a balance does exist between the number of exposures required to improve physical conditioning and the number that elicits negative fatigue responses and increased injury rates. The sole investigation into workload and injury in AmF found athletes who are able to produce high average inertial load values were less susceptible to contact and non-contact musculoskeletal strains and sprains [22]. Players with a high level of neuromuscular responsiveness and control are able to produce high inertial load values through rapid acceleration, deceleration and change of direction, which may offer protective benefits. However, if an athlete is fatigued, their level of neuromuscular control decreases and may increase risk of injury [22].

To increase and maintain physical capacities such as muscular strength and power, AmF athletes complete multiple resistance training sessions each week in combination with field-based skill training [1]. This is similar to other contact sports such as rugby league where resistance training loads were significantly correlated with both contact ($r = 0.75$) and non-contact ($r = 0.82$) field injuries [17]. This may be due to resistance and field training scheduling issues that did not allow the athletes to recover adequately prior to the following field session or inadequate athlete monitoring strategies that did not identify the training readiness of the athlete. It is suggested that the multidisciplinary team of practitioners and coaches ensure resistance training and field sessions are carefully planned in order to reduce carryover of residual fatigue and muscle soreness into the following session as inadequate recovery may result in negative training adaptations and injury [26]. The following evidence-based principles can be implemented into routine athlete monitoring and load management to form a targeted injury prevention approach in AmF that may assist practitioners with monitoring the collision- and movement-based workload to minimise fatiguing effects such as injury [24].

6.1 Establishing a Moderate Chronic Workload

The risk of injury increases with both excessively high workloads (overload) and low workloads (underloading), indicating a 'U-shaped relationship' [12, 14, 46]. Moderate workloads, however, have been reported to be protective against injury in rugby union, with a monthly internal workload of approximately 6000–8500 arbitrary units (AU) reducing the risk of injury [12]. Similarly, the number of exposures to maximal velocity that may offer a 'counter measure' against subsequent injury is related to the individual's chronic training load [46]. A chronic workload of >4750 AU allowed athletes to perform a greater number of maximum-velocity exposures, offering a protective effect against injury compared with low chronic workloads (<4750 AU) who were more susceptible to non-contact injury after completing a similar number of exposures to maximum velocity [46]. To reduce the injury risk in AmF athletes who are required to compete at maximum velocities, it is recommended that coaches prepare well-planned, physically demanding training to provide high chronic training loads. This allows athletes to be appropriately exposed to greater number of maximum-velocity efforts that will adequately prepare them for maximum-velocity exposures in competition without subsequent injury.

6.2 Minimise Week-to-Week Changes in Workload

Weekly spikes in workload are associated with increased injury risk [12, 20, 47], with athletes at risk for up to 1 month after a spike [48]. In ARF, rapid changes (>10%) in weekly workload were associated with 40% of injuries in the following week [47]. Weekly changes in internal load of >1250 AU and 1069 AU were also linked to increased injury risk in ARF [20] and rugby union [12], respectively. Players with a constant week-to-week workload increase of up to a 10% had <10% risk of injury, whereas when workload increased by >15% the risk of injury increased up to 49% [16]. It is recommended that practitioners attempt to limit weekly increases in workload to <10%. If a spike in workload is identified, careful workload management is required during the subsequent month to ensure that the effect of the spike does not result in injury [16, 24].

6.3 Maintain Minimum Workloads

In AmF, a reduction in chronic workload through holidays or post-season breaks may require appropriate load management upon return to training to avoid spikes in workload. For example, in collegiate AmF, athletes are provided a break at the end of the spring training period before the summer holidays. The spring training period develops an athlete's chronic workload that may protect them against injury; however, during the multiple-week break the athletes are under no obligation to train and staff are not permitted to monitor any training the athlete may complete [49]. This has been termed the out-of-season 'black hole', where unfortunately many athletes return to training considerably deconditioned, thus increasing the risk of injury [49]. Upon return it is important that coaches and practitioners consider the chronic workload of the athletes and prescribe training accordingly to avoid spikes in acute workload [12, 14, 16, 20]. However, it should also be the responsibility of the athlete to ensure a minimum workload is maintained throughout pre-planned breaks such as the off season and holiday periods as everyone is involved in the training process and successful team output at various levels [26].

6.4 Avoid the Maximum Workload

Weekly and monthly accumulated external and internal workload thresholds have been identified and associated with injury in several contact sports [12, 18, 20, 44]. In ARF, 3-weekly distance (73,721–86, 662 m), sprint distance (864–1453 m) and force load (>5397 m) were significantly linked with injury [44]. Whilst in rugby league the risk of injury was 2.7 times greater when sprinting (>7 m/s) surpassed 9 m per session [18]. Accumulated weekly and 14-day internal loads of >1750 and >4000 AU, respectively, have been linked with an increase in injury risk in ARF [20], whilst weekly and monthly workloads of >1245 and >8651 AU, respectively, are associated with a substantial increase in the risk of injury in rugby union [12]. Whilst it is difficult to make meaningful comparisons of the physiological demands within and across team sports due to a multitude of factors, it is recommended that practitioners conduct internal investigations to identify the ceiling effect within their team [4, 7].

6.5 Managing Competition Loads

It is common during the season that athletes may decondition through experiencing lower training and/or competition loads often seen in reserve and non-selected players [50], moving them from a moderate to low chronic workload and increasing the risk of injury [12, 16, 24]. This presents a problem in AmF due to the number of players in a squad who may not be selected to participate in many plays. In this instance early identification and modification (adding load) of workloads is warranted in order to maintain a moderate workload [12, 24]. In contrast, if an athlete has spiked in the number of plays they are involved in their workload needs to be managed appropriately to ensure that the spike does not lead to subsequent injury [12, 24].

7 Conclusion

The sport of AmF requires highly developed physical capacities of muscular strength, power and speed along with the ability to perform position-specific technical skills [1]. In competition athletes are frequently performing at maximal intensity and velocity. They are also exposed to a large number and magnitude of collisions through tackling and blocking. In order to prepare athletes for the rigours of competition they need to be exposed to a training stimulus that can elicit positive physiological adaptations that can improve these physical capacities. However, if the training stimulus and recovery status of individual athletes are not planned and monitored adequately, negative training adaptations such as injury may occur. To date, only one study has investigated the link between workload and injury in AmF, which reported high average inertial load values with decreased risk of injury [22]. Higher inertial load values are generated through rapid acceleration, deceleration and change of direction, which may be associated with increased neuromuscular control and responsiveness [22]. This suggests that negative fatigue effects such as decreased neuromuscular control through inadequate workload monitoring and management may increase

the risk of injury in AmF [22]. Adapting and implementing evidence-based workload monitoring practices from other team-based contact sports may decrease injury risk, ensure athlete safety and maximise team success. This requires a multidisciplinary approach between coaches, strength and conditioning staff, sport scientists, athletic trainers and medical staff as each role is somewhat involved in the training process. Each member of the coaching and performance staff needs to have an appreciation of their own role and the role that others play to reach the same goal of maximising on-field competitive performance.

Compliance with Ethical Standards

Funding No financial support was obtained for this study.

Conflict of interest Toby Edwards, Tania Spiteri, Benjamin Piggott, G. Gregory Haff and Christopher Joyce declare that they have no conflicts of interest relevant to the content of this review article.

References

1. Hoffman JR. The applied physiology of American Football. Int J Sport Physiol Perform. 2008;3:387–92.
2. Wellman AD, Coad SC, Goulet GC, Coffey VG, McLellan CP. Quantification of accelerometer derived impacts associated with competitive games in NCAA division I college football players. J Strength Cond Res. 2016;31(2):330–8.
3. Wellman AD, Coad SC, Goulet GC, McLellan CP. Quantification of competitive game demands of NCAA division I college football players using global positioning systems. J Strength Cond Res. 2016;30(1):11–9.
4. Cummins C, Orr R, O'Connor H, West C. Global positioning systems (GPS) and microtechnology sensors in team sports: a systematic review. Sports Med. 2013;43(10):1025–42.
5. Gabbett TJ. Quantifying the physical demands of collision sports: does microsensor technology measure what it claims to measure? J Strength Cond Res. 2013;27(8):2319–22.
6. Gabbett TJ, Jenkins DG, Abernethy B. Physical demands of professional rugby league training and competition using microtechnology. J Sci Med Sport. 2012;15(1):80–6.
7. Malone JJ, Lovell R, Varley MC, Coutts AJ. Unpacking the black box: applications and considerations for using GPS devices in sport. Int J Sports Physiol Perform. 2017;12(2):18–26.
8. McLean BD, Coutts AJ, Kelly V, McGuigan MR, Cormack SJ. Neuromuscular, endocrine, and perceptual fatigue responses during different length between-match microcycles in professional rugby league players. Int J Sport Physiol Perform. 2010;5(3):367–83.
9. McLellan CP, Lovell DI, Gass GC. Biochemical and endocrine responses to impact and collision during elite rugby league match play. J Strength Cond Res. 2011;25(6):1553–62.
10. Dick R, Ferrara MS, Agel J, Courson R, Marshall SW, Hanley MJ, et al. Descriptive epidemiology of collegiate men's football injuries: National Collegiate Athletic Association Injury Surveillance System, 1988–1989 through 2003–2004. J Athl Train. 2007;42(2):221–33.
11. Coutts AJ, Reaburn P, Piva TJ, Rowsell GJ. Monitoring for overreaching in rugby league players. Eur J Appl Physiol. 2007;99(3):313–24.
12. Cross MJ, Williams S, Trewartha G, Kemp SP, Stokes KA. The influence of in-season training loads on injury risk in professional rugby union. Int J Sports Physiol Perform. 2016;11(3):350–5.
13. Drew M, Purdam C. Time to bin the term 'overuse' injury: is 'training load error' a more accurate term? Br J Sports Med. 2016;50:1423–4.
14. Drew MK, Finch CF (2016) The relationship between training load and injury, illness and soreness: a systematic and literature review. Sports Med. 2016;46(6):861–83.
15. Gabbett T, Jenkins D, Abernethy B. Physical collisions and injury during professional rugby league skills training. J Sci Med Sport. 2010;13(6):578–83.
16. Gabbett TJ. The training—injury prevention paradox: should athletes be training smarter and harder? Br J Sports Med. 2016;50(5):273–80.
17. Gabbett TJ, Jenkins DG. Relationship between training load and injury in professional rugby league players. J Sci Med Sport. 2011;14(3):204–9.
18. Gabbett TJ, Ullah S. Relationship between running loads and soft-tissue injury in elite team sport athletes. J Strength Cond Res. 2012;26(4):953–60.
19. Hulin BT, Gabbett TJ, Lawson DW, Caputi P, Sampson JA. The acute: chronic workload ratio predicts injury: high chronic workload may decrease injury risk in elite rugby league players. Br J Sports Med. 2015;50:231–6.
20. Rogalski B, Dawson B, Heasman J, Gabbett TJ. Training and game loads and injury risk in elite Australian footballers. J Sci Med Sport. 2013;16(6):499–503.
21. Veugelers KR, Young WB, Fahrner B, Harvey JT. Different methods of training load quantification and their relationship to injury and illness in elite Australian football. J Sci Med Sport. 2016;19(1):24–8.
22. Wilkerson GB, Gupta A, Allen JR, Keith CM, Colston MA. Utilization of practice session average inertial load to quantify college football injury risk. J Strength Cond Res. 2016;30(9):2369–74.
23. DeMartini JK, Martschinske JL, Casa DJ, Lopez RM, Ganio MS, Walz SM, et al. Physical demands of national collegiate athletic association division I football players during preseason training in the heat. J Strength Cond Res. 2011;25(11):2935–43.
24. Drew MK, Cook J, Finch CF. Sports-related workload and injury risk: simply knowing the risks will not prevent injuries: narrative review. Br J Sports Med. 2016;50:1306–8.
25. Windt J, Gabbett TJ. How do training and competition workloads relate to injury? The workload–injury aetiology model. Br J Sports Med. 2017;51(5):428–35.
26. Mooney M, Charlton PC, Soltanzadeh S, Drew MK. Who 'owns' the injury or illness? Who 'owns' performance? Applying systems thinking to integrate health and performance in elite sport. Br J Sports Med. 2017;51(14):1054–5.
27. Fry AC, Kraemer WJ. Physical performance characteristics of american collegiate football players. J Strength Cond Res. 1991;5(3):126–38.
28. Garstecki MA, Latin RW, Cuppett MM. Comparison of selected physical fitness and performance variables between NCAA Division I and II football players. J Strength Cond Res. 2004;18(2):292–7.
29. Akubat I, Barrett Abt G. Integrating the internal and external training loads in soccer. Int J Sports Physiol Perform. 2014;9(3):457–62.
30. Boyd LJ, Ball K, Aughey RJ. The reliability of MinimaxX accelerometers for measuring physical activity in Australian football. Int J Sports Physiol Perform. 2011;6(3):311–21.
31. Hulin BT, Gabbett TJ, Johnston RD, Jenkins DG. Wearable microtechnology can accurately identify collision events during

professional rugby league match-play. J Sci Med Sport. 2017;20(7):638–42.
32. Coughlan GF, Green BS, Pook PT, Toolan E. O'connor SP. Physical game demands in elite rugby union: a global positioning system analysis and possible implications for rehabilitation. J Orthop Sports Phys Therp. 2011;41(8):600–5.
33. Cunniffe B, Proctor W, Baker JS, Davies B. An evaluation of the physiological demands of elite rugby union using global positioning system tracking software. J Strength Cond Res. 2009;23(4):1195–203.
34. McLellan CP, Lovell DI. Neuromuscular responses to impact and collision during elite rugby league match play. J Strength Cond Res. 2012;26(5):1431–40.
35. Iosia MF, Bishop PA. Analysis of exercise-to-rest ratios during division IA televised football competition. J Strength Cond Res. 2008;22(2):332–40.
36. Feeley BT, Kennelly S, Barnes RP, Muller MS, Kelly BT, Rodeo SA, et al. Epidemiology of National Football League training camp injuries from 1998 to 2007. Am J Sports Med. 2008;36(8):1597–603.
37. Shankar PR, Fields SK, Collins CL, Dick RW, Comstock RD. Epidemiology of high school and collegiate football injuries in the United States, 2005–2006. Am J Sports Med. 2007;35(8):1295–303.
38. Willigenburg NW, Borchers JR, Quincy R, Kaeding CC, Hewett TE. Comparison of injuries in American collegiate football and club rugby: a prospective cohort study. Am J Sports Med. 2016;44(3):753–60.
39. Benson BW, McIntosh AS, Maddocks D, Herring SA, Raftery M, Dvorak J. What are the most effective risk-reduction strategies in sport concussion? Br J Sports Med. 2013;47(5):321–6.
40. Black A, Palacios-Derflingher L, Schneider KJ, Hagel BE, Emery CA. The effect of a national body checking policy change on concussion risk in youth ice hockey players. Br J Sports Med. 2017;51(11):A70–1.
41. Meeusen R, Duclos M, Foster C, Fry A, Gleeson M, Nieman D, et al. Prevention, diagnosis, and treatment of the overtraining syndrome: joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. Med Sci Sports Exerc. 2013;45(1):186–205.
42. Gabbett TJ. Changes in physiological and anthropometric characteristics of rugby league players during a competitive season. J Strength Cond Res. 2005;19(2):400–8.
43. Gallo T, Cormack S, Gabbett T, Williams M, Lorenzen C. Characteristics impacting on session rating of perceived exertion training load in Australian footballers. J Sports Sci. 2015;33(5):467–75.
44. Colby MJ, Dawson B, Heasman J, Rogalski B, Gabbett TJ. Accelerometer and GPS-derived running loads and injury risk in elite Australian footballers. J Strength Cond Res. 2014;28(8):2244–52.
45. Gabbett TJ. Reductions in pre-season training loads reduce training injury rates in rugby league players. Br J Sports Med. 2004;38(6):743–9.
46. Malone S, Roe M, Doran DA, Gabbett TJ, Collins K. High chronic training loads and exposure to bouts of maximal velocity running reduce injury risk in elite Gaelic football. J Sci Med Sport. 2016;20(3):250–4.
47. Piggott B, Newton MJ, McGuigan MR. The relationship between training load and incidence of injury and illness over a pre-season at an Australian Football League club. J Aust Strength Cond. 2009;17:4–17.
48. Orchard JW, James T, Portus M, Kountouris A, Dennis R. Fast bowlers in cricket demonstrate up to 3-to 4-week delay between high workloads and increased risk of injury. Am J Sports Med. 2009;37(6):1186–92.
49. Gabbett TJ, Kennelly S, Sheehan J, Hawkins R, Milsom J, King E, Whiteley R, Ekstrand J. If overuse injury is a 'training load error', should undertraining be viewed the same way? Br J Sports Med. 2016;50:1017–8.
50. Anderson L, Orme P, Michele RD, Close GL, Milsom J, Morgans R, et al. Quantification of seasonal-long physical load in soccer players with different starting status from the English Premier League: implications for maintaining squad physical fitness. Int J Sports Physiol Perform. 2016;11(8):1038–46.